

EVIDENCE FOR THE FUNDAMENTAL DIFFERENCE HYPOTHESIS OR NOT?

Island Constraints Revisited

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This article examines how changes in linguistic theory affect the debate between the fundamental difference hypothesis and the access-to-Universal Grammar (UG) approach to SLA. With a focus on subadjacency (Chomsky, 1973), a principle of UG that places constraints on *wh*-movement and that has frequently been taken as a test case for verifying second language (L2) access to UG, we reanalyze earlier L2 findings in terms of a revised constraint, which effectively prohibits extraction out of subjects and adjuncts. We show that L2 learners indeed observe such a constraint on *wh*-movement, and, at the same time, we suggest that recent claims for a universal computational system (e.g., Chomsky, 1995; Uriagereka, 1999) make the respective roles of the first language and UG difficult to tease apart.

According to the fundamental difference hypothesis (FDH; Bley-Vroman, 1989, 1990), adult SLA differs from child first language (L1) acquisition in a number of respects. These differences are attributed to the non-availability of domain-specific language acquisition mechanisms and Universal Grammar (UG) for adult second language (L2) learning, in contrast to child language acquisition. An implication of this approach is that the outcome of SLA, the interlanguage grammar, is not inevitably UG-constrained. However, recognizing that adult L2 learners are often successful in the acquisition of subtle properties underdetermined by

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the input and not explicitly taught (i.e., assumed to derive from UG), the FDH posits that this kind of knowledge must come from the L1. In other words, the native language exemplifies linguistic universals (by definition) and can be used, in some sense, to reconstruct some aspects of—or build a surrogate for—UG (Bley-Vroman, 1989). Such reconstruction is necessarily restricted to “the most obvious large-scale characteristics of the native language” (Bley-Vroman, 1989, p. 52).

In part as a response to this line of argument, researchers (e.g., White, 1985, 1988) who sought to demonstrate that UG is in fact available (in addition to supporters of the FDH [e.g., Schachter, 1989, 1990], who had the opposite goal in mind) conducted studies in which two conditions were held constant (White, 1990): (a) The phenomenon under investigation constituted a learnability or poverty-of-the-stimulus problem in that it could not be readily induced from the L2 input or learned on the basis of instruction and thus motivated an explanation in terms of UG; (b) the universal principle under investigation did not operate in the L1, or the L1 and L2 differed with respect to parameter settings. Under these two conditions, the FDH predicts failure (i.e., principles of UG will be violated; L2 parameter settings will not be acquired), whereas proponents of UG access predict success (i.e., principles of UG will be observed; L2 parameter settings will be acquired).

Along these lines, constraints on *wh*-movement have frequently been adopted as a test case to verify whether interlanguage grammars are UG-constrained. Several L2 studies investigated knowledge of the subadjacency condition (Chomsky, 1973) in the interlanguage grammars of L2 learners whose L1s leave *wh*-phrases in their thematic position and thus apparently lack *wh*-movement. In other words, these learners could not rely on the L1 to construct adequate surrogates for the relevant UG constraints on *wh*-movement. Results in some cases suggest that adult L2 learners are quite inaccurate at rejecting subadjacency violations, which supports claims for the nonavailability of UG—in other words, the FDH (e.g., Johnson & Newport, 1991; Schachter, 1989, 1990)¹—whereas other studies report success as far as subadjacency is concerned (e.g., White & Juffs, 1998).

In the debate for or against UG in SLA, arguments are dependent on the presumed nature of UG itself. Attempts to account adequately for subadjacency effects mirror the development of linguistic thought over the years: Linguists originally identified a set of isolated conditions (Chomsky, 1964; Ross, 1967) and then moved toward a more unified and principled account (e.g., Uriagereka, 1999). Such changes in linguistic theory may necessitate the revision of assumptions about the role of the L1 and what L2 learners can or cannot acquire. Here, we discuss changes in theoretical accounts of how *wh*-movement is constrained and implications of these changes for theories of SLA.

Discussion of the implications for L2 research of ongoing changes to linguistic theory is not new. Poverty-of-the-stimulus problems crucially motivate a role for UG and remain constant regardless of changes in the formulation of principles proposed to account for acquisition (e.g., Schwartz & Sprouse, 2000; White, 1995). This is the case for constraints on *wh*-movement. At the same time, several researchers (Dekydtspotter, Sprouse, & Anderson, 1997; Hale, 1996; White, 1995) have noted that, under Minimalism (Chomsky, 1995), computational principles such as *move* and *merge* are assumed to be invariant (therefore, universal) and locality phenomena are exemplified in all languages, making it difficult—if not impossible—to disentangle L1 effects from UG effects. In the case of *wh*-movement, we will conclude that even when L2 learners are found to be successful in surmounting learnability problems, it may be impossible to determine whether such success is attributable to UG alone, to the L1 alone, or to some combination of the two. However, although the FDH and UG-access accounts seem to overlap in their attribution of a role to the L1, these approaches can be distinguished based on the nature of L1 transfer allowed by each.

ISLANDS VERSION 1: ISLANDS AND CONSTRAINTS

Wh-movement in English has long been observed not to be free but, rather, circumscribed or bounded (bounding theory; Chomsky, 1981). Certain circumstances have been identified in which *wh*-movement can take place and others in which it is prohibited, as illustrated in (1)–(7). Examples (1b) and (2b) are grammatical: *Wh*-movement takes place from a matrix clause and an embedded clause, respectively. In contrast, the ungrammatical sentences in (3b)–(7b) involve extraction from a variety of embedded constituents, referred to as islands (Ross, 1967).

- (1) a. *This girl danced with Mark.*
b. *Who did this girl dance with ____?*
- (2) a. *You said that this girl danced with Mark.*
b. *Who did you say that this girl danced with ____?*
- (3) a. *You wondered whether this girl danced with Mark.*
b. **Who did you wonder whether this girl danced with ____?*
- (4) a. *You spread a rumor that this girl danced with Mark.*
b. **Who did you spread a rumor that this girl danced with ____?*
- (5) a. *You met a girl that danced with Mark.*
b. **Who did you meet a girl that danced with ____?*

- (6) a. *That this girl danced with Mark annoyed you.*
 b. **Who did that this girl danced with ____ annoy you?*
- (7) a. *You met this girl after she danced with Mark.*
 b. **Who did you meet this girl after she danced with ____?*

Acquisition of knowledge of this kind of ungrammaticality, it is assumed, constitutes a learnability problem: The relevant information is not available in the naturalistic input to which the language learner is exposed. L1 (or L2) learners will come across numerous examples of grammatical *wh*-movement and could reasonably infer (contrary to fact) that *wh*-movement is free. Instead, language acquirers must come to know that, in certain situations, movement is not possible.

Early explanations of learners' avoidance of such ungrammaticality were couched in terms of built-in, or innate, principles: Universal constraints that the learner does not have to acquire assist in the acquisition of *wh*-movement. Proposed constraints included the following: (a) the *wh*-island constraint (Chomsky, 1964), which prohibits the movement of *wh*-elements out of an embedded clause introduced by another *wh*-phrase, as illustrated in (3b); (b) the complex noun phrase (NP) constraint (Ross, 1967), which bans extraction out of the sentential complements of nouns (Ns) or relative clauses, as illustrated in (4b) and (5b), respectively; (c) the subject condition (Chomsky, 1973), which excludes extraction from a subject, as illustrated in (6b); and (d) the adjunct island condition (Huang, 1982a), which prohibits extraction out of an adjunct phrase, as illustrated in (7b). The effect of these constraints is to stipulate that certain constituents form islands from which *wh*-phrases cannot escape. Such disparate constraints were brought together under the subadjacency condition (Chomsky, 1973), a simplified version of which is provided in (8). The rationale for this principle was to capture the different kinds of islands in a uniform way.

- (8) The subadjacency condition: A constituent may not move over more than one bounding category at a time.

Bounding nodes were originally proposed to be *sentence* (S) and NP (now inflection phrase [IP] and determiner phrase [DP], respectively). As an illustration of how the subadjacency condition predicts the contrast between the grammatical sentence illustrated in (2b) and the ungrammatical sentence illustrated in (4b), these sentences are repeated in (9) with the relevant bracketing (with trace [*t*] marking the various positions of the *wh*-phrase in the course of the derivation). Crucially, long-distance movement must be cyclic; it takes place by means of a series of shorter, local movements. In (9a), *wh*-movement crosses just one bounding node (IP) at a time, and the sentence

phrase [Spec, CP] for feature-checking purposes. In [-*wh* movement] languages, the Q-feature is weak, which results in covert movement at logical form (LF).

Second language learners of English who are native speakers of *wh*-in-situ languages are faced with the same underdetermination problem as children who acquire English as a L1: The English input will include examples of grammatical cases of movement but no information about ungrammaticality.⁴ L2 learners, then, have to acquire subtle phenomena (i.e., constraints on *wh*-movement) not obvious from the L2 input and not derivable from the L1, a situation that meets the two requirements for the evaluation of the FDH against the UG-access approach.⁵

Although the exact formulation of the relevant constraint(s) has been constantly changing, the idea has always been that such a constraint should be universal, even though languages that lack *wh*-movement, such as Chinese, Japanese, or Korean, presumably do not make use of it. Consequently, if interlanguage grammars are constrained by UG, the effects of subjacency are predicted to be observed in the grammars of L2 learners acquiring a *wh*-movement language like English, regardless of whether their L1s employ *wh*-movement. L2 learners, then, should recognize the ungrammaticality of sentences that violate subjacency. In other words, if UG is still effective in SLA, once learners realize that the L2 employs *wh*-movement, they should be able to reject subjacency violations, in spite of the apparent learnability problem and the differences between the L1 and the L2 with respect to *wh*-constructions.

In contrast, according to the FDH, L2 learners whose L1 lacks *wh*-movement will not have access to UG (either directly or indirectly) and thus are expected to fail to observe constraints on *wh*-movement. In other words, subjacency violations should be accepted as grammatical by L2 learners, which would indicate that their grammars do not fall within the range determined by UG (i.e., that interlanguages are not natural languages).

Earlier L2 studies that adopted this kind of logic and assumed this version of island constraints include Bley-Vroman et al. (1988), Johnson and Newport (1991), Li (1998),⁶ Schachter (1989, 1990), and White and Juffs (1998), among others.⁷ These studies, summarized in Table 1, have several common themes: (a) They include L2 learners with *wh*-in-situ L1s;⁸ (b) they investigate several of the island constraints subsumed under subjacency; and (c) they generally demonstrate that, overall, L2 performance on subjacency violations is less accurate than performance on equivalent grammatical sentences as well as less accurate than native-speaker performance, which arguably supports the FDH (but see Martohardjono, 1993). However, learners in these studies treat the various islands differently: It is not the case that all violations are recognized equally.

Table 1. Comparison of previous L2 studies

Study	L1s	Island type	Results on violations
Bley-Vroman et al. (1988)	Korean	<i>Wh</i> -islands, N complements, RCs	Accurate on <i>wh</i> -islands (87%) and RCs (84%), less accurate on N complements (65%).
Johnson and Newport (1991)	Chinese	<i>Wh</i> -islands, N complements, RCs	Accurate on RCs (79%), at chance on N complements (46%) and <i>wh</i> -islands (50%).
Li (1998)	Chinese	<i>Wh</i> -islands, RCs, SSs	More accurate on RCs (76%) and SSs (70%) than on <i>wh</i> -islands (52%).
Schachter (1989, 1990)	Chinese, Korean	<i>Wh</i> -islands, N complements, RCs, SSs	More accurate on RCs (56%) and SSs (63%) than on N complements (43%) and <i>wh</i> -islands (43%). ^a
White and Juffs (1998)	Chinese	<i>Wh</i> -islands, N complements, RCs, SSs, adjunct islands	Accuracy on all island types was 84% or higher; no differences were found between them. ^b

Note. RC = relative clause; SS = sentential subject.

^a These results are from the Korean subjects in the 1990 study.

^b These results are from the group of Chinese speakers tested in China.

In sum, considerable variation is found in the degree of accuracy with which L2 learners recognize subjacency violations. However, this variation can, in part, be attributed to methodological differences.⁹ On the basis of their results, Schachter (1989, 1990) and Johnson and Newport (1991) concluded that adult L2 learners do not have (full) access to UG: Learners acquire *wh*-movement but fail to observe the constraints that govern it; that is, these results support the FDH. Bley-Vroman et al. (1988) were more cautious and argued that their learners performed at a level much higher than chance, which suggests, possibly, that UG operates in an attenuated form. In contrast, White and Juffs (1998) and Li (1998) argued that their results support the claim that interlanguage grammars are constrained by UG.

One line of response to results that report failure to reject violations (e.g., White, 1992) has been to point out that learners may not

have acquired *wh*-movement (yet). Before it can be determined whether island constraints are operative in the interlanguage grammar, it is necessary to establish that *wh*-movement has in fact been acquired; that is, the [$\pm wh$ movement] parameter has to be reset before the principle can come into play. Most of the studies summarized in Table 1 were conducted under the assumption that the [$+wh$ movement] value had been acquired; in some cases, grammatical *wh*-questions were included to establish this point. However, it is conceivable that learners might form English *wh*-questions without *wh*-movement. Two different analyses of this type have been proposed. One analysis suggested that L2 learners form *wh*-questions by means of topicalization of a *wh*-phrase, with the topic linked to a null pronoun, which conforms to the topicalization possibilities allowed by their L1s (Hawkins & Chan, 1997; Martohardjono & Gair, 1993; White, 1992). An alternate analysis suggested that L2 learners form English *wh*-questions by means of scrambling, a possibility allowed in Japanese (Hawkins, 2001, 2005; Hawkins & Hattori, 2006).¹⁰ The Japanese examples in (11) show the contrast between *wh*-in-situ, in which the *wh*-phrase *nani* “what” stays in its thematic position, and scrambling, as illustrated in (11a) and (11b), respectively. The fact that scrambling is not subject to subjacency explains why L2 learners produce what look like English *wh*-questions but fail to reject subjacency violations.¹¹

- (11) a. *John-ga kinou nani-o kaimashita ka?*
 John-NOM yesterday what-ACC buy-PAST Q
 “What did John buy yesterday?”
- b. *nani-o John-ga kinou kaimashita ka?*
 what-ACC John-NOM yesterday buy-PAST Q
 “What did John buy yesterday?” (Hawkins, 2005, p. 130)

Umeda (2005, 2006) argued against this approach through the investigation of related phenomena to determine whether L1 Japanese learners represent English *wh*-questions in terms of true *wh*-movement, topicalization, or scrambling. She looked at three phenomena associated with true *wh*-movement—namely, scope freezing, reconstruction, and weak crossover. Topicalization in Japanese does not show reconstruction or weak crossover effects, whereas scrambling does not show scope freezing or weak crossover. Umeda (2006) found that L2 learners display knowledge of scope freezing, reconstruction, and weak crossover, which suggests true *wh*-movement. Moreover, Umeda (2005) found that although a minority of learners exhibited behavior consistent with a topicalization analysis, no learners behaved in a manner consistent with scrambling.

Next, we will focus on the inconsistencies found across the different types of subjacency violations. Such inconsistencies are not predicted by either the FDH or the UG-access approach. Under the original formulation of subjacency, all violations should be treated in the same way. In other words, learners should either recognize all of the violations or none of them. These inconsistencies are better understood under subsequent formulations of island constraints.

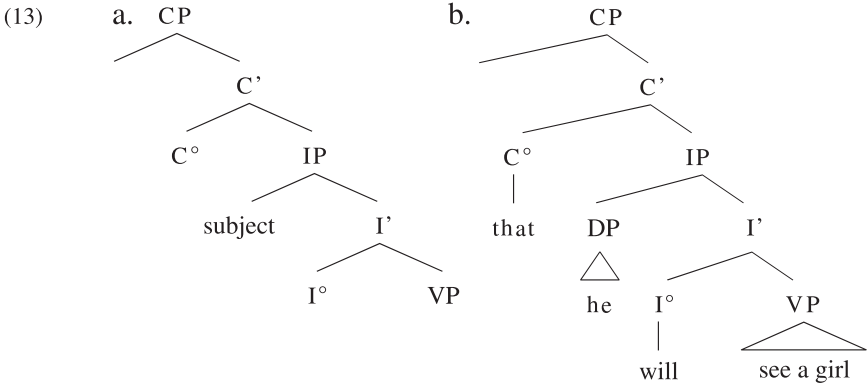
ISLANDS VERSION 2: CONDITIONS ON EXTRACTION DOMAINS AND BARRIERS

The subjacency condition, as formulated by Chomsky (1973), turned out to be both too strong and too weak; for example, some sentences that should be ruled out are grammatical, and some sentences that are ungrammatical are not ruled out. In the case of (12a), the sentence is grammatical even though a DP and an IP have been crossed; in the case of (12b), the sentence is ungrammatical—involving extraction from an adjunct—but only one bounding node has been crossed (*viz.* the IP).

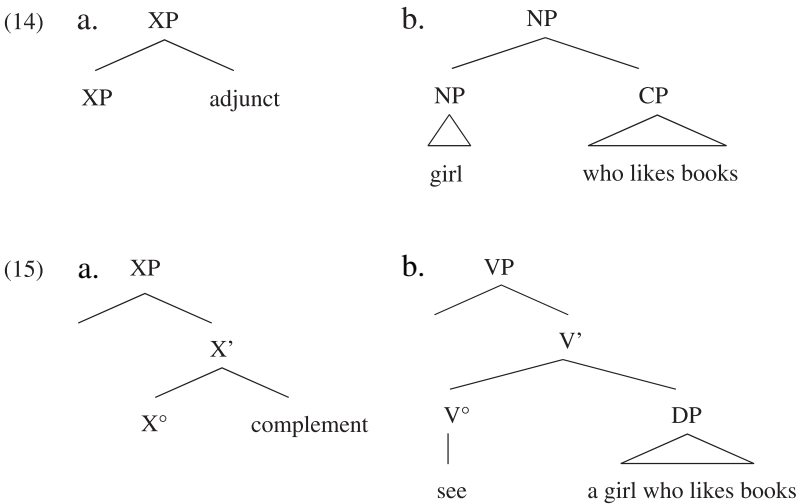
- (12) a. [*Which city*]_i *did* IP[*you witness* DP[*the destruction of t_i*]]?
 b. *[*Which city*]_i *did* IP[*you visit Paris* PP[*before t_i*]]?

Furthermore, crosslinguistic differences were uncovered; for instance, extractions from *wh*-islands are acceptable in languages like Italian, French, and Hebrew. In consequence, it was proposed that there must be parametric differences in bounding nodes, with some languages having S' (now CP) as a bounding node instead of S (or IP; Rizzi, 1982; Sportiche, 1981); alternatively, an extra landing site was proposed (Reinhart, 1981). In short, the subjacency condition as formulated by Chomsky (1973) turned out to be far from universal; it held neither crosslinguistically nor for all extractions in English. As a result, revisions were proposed to account for island phenomena.

Another attempt to unify some of the original island constraints was the condition on extraction domains (CED) proposed by Huang (1982a; see also Cattell, 1976), which united the subject condition and the adjunct island condition, on the basis of the fact that subjects and adjuncts are not governed by lexical categories.¹² In brief, a lexical head governs a phrase if no maximal projection (*i.e.*, phrasal node) intervenes between them. Accordingly, subjects, which occur as constituents in [Spec, IP], are not lexically governed because the CP normally blocks government by a lexical head from above, whereas the verb phrase (VP) blocks it from below, as illustrated in (13). Similarly, adjuncts are not lexically governed because they are adjoined; that is,



they are positioned between two layers of the same maximal projection, as illustrated in (14). In other words, the CED approach distinguishes between complements, which are necessarily governed by the lexical head that assigns them a thematic role (theta-marking), as illustrated in (15), and noncomplements, which are the only true islands. The trees in (13a), (14a), and (15a) show the relevant structural configurations; (13b), (14b), and (15b) illustrate how such configurations can be implemented.



Few L2 studies have been conducted with the CED as the principle under investigation (but see Martohardjono, 1993); however, many of Huang's (1982a) insights were incorporated by Chomsky (1986a) into the *Barriers* framework.

The system described in *Barriers* (Chomsky, 1986a) was one attempt to overcome some of the shortcomings of earlier formulations and to explain various different subtheories of grammar—such as the empty category principle (ECP) and subjacency—in the same way.¹³ One consequence was a revised definition of subjacency, which involved barriers and blocking categories rather than bounding nodes and incorporated the notion of government as well as the idea of degrees of grammaticality. Barriers are relative rather than absolute, in contrast to bounding nodes. Every maximal projection is a potential barrier depending on the circumstances: To become an actual barrier, the maximal projection either must not be theta-governed (i.e., a blocking category) or must be the first maximal projection to dominate a blocking category (hence, a barrier by inheritance). A distinction was made between stronger and weaker islands, determined by the number of barriers that had to be crossed to escape from the island: Crossing one barrier gave rise to a weak subjacency violation, whereas crossing a barrier along with a violation of the ECP or crossing two barriers caused a greater reduction in acceptability, as in (16). Extractions out of subjects and adjuncts always result in strong violations, whereas weak islands are complements (of verbs [Vs] or Ns).

- (16) a. *Strong islands: subjects, adjuncts (including relative clauses)*
 b. *Weak islands: wh-islands, N complements*¹⁴

Under the *Barriers* system (Chomsky, 1986a), the extraction in (12b), repeated in (17b), arguably crosses two barriers. The prepositional phrase (PP) *before which city* is an adjunct and thus a barrier; under the assumption that adjuncts are adjoined to VP, IP is the second barrier (by inheritance). In contrast, the extraction in (12a), repeated in (17a), does not cross any barrier because the NP *the destruction of which city* is theta-governed by the V *witness*.

- (17) a. [*Which city*]_i *did* IP[*you witness* DP[*the destruction of t_i*]]?
 b. *[*Which city*]_i *did* IP[*you visit Paris* PP[*before t_i*]]?

This distinction between strong and weak islands allowed a fresh look at what might be expected in SLA given that L2 learners do not treat all subjacency violations alike. They are more accurate at recognizing some types of violations than others: For the most part, strong violations are recognized as such, whereas weak ones may not be. Johnson and Newport's (1991) L2 learners performed above chance on extractions out of relative clauses (which are strong violations) and at chance on extractions out of noun complements and *wh*-islands (which are weak violations). Although they were quite inaccurate overall, the L2 learners studied by Schachter (1989, 1990) were better at rejecting

extractions out of relative clauses and subjects (i.e., strong violations) than out of noun complements and *wh*-islands (i.e., weak violations). Bley-Vroman et al.'s (1988) L2 learners were more accurate on ungrammatical extractions from relative clauses (i.e., strong violations) than from noun complements (i.e., weak violations). It is interesting to note that these learners were also accurate on *wh*-islands, which are weak violations. Li's (1998) L2 learners were more accurate on ungrammatical extractions from relative clauses and sentential subjects (i.e., strong violations) than from *wh*-islands (i.e., weak violations). In the case of White and Juff's (1998) study, L2 learners generally performed more accurately than learners in the other studies: Almost all violations tested were strong. Additionally, Hawkins and Chan (1997) reported that their Chinese speakers were inaccurate on subjacency violations, although this can be explained by the fact that their test sentences were restricted to extractions out of *wh*-islands and noun complements (i.e., weak violations). Similarly, those sentences relevant to subjacency in the study by Hawkins and Hattori (2006) involved *wh*-island violations only.

Indeed, Martohardjono (1993) showed that L2 learners' differential treatment of various types of ungrammatical extractions falls out from Chomsky's (1986a) *Barriers* system. She argued that if L2 learners exhibit a higher rate of rejection for stronger violations than for weaker violations, such linguistic behavior must stem from their knowledge of the relevant UG principles. Martohardjono provided further support for the claim that L2 learners are more sensitive to strong violations (i.e., relative clauses, clausal adjuncts, sentential subjects) than weak (i.e., sentential complements of nouns, *wh*-islands). She tested L2 learners whose native languages differed with respect to syntactic movement: Indonesian (no syntactic movement, according to Martohardjono), Chinese (syntactic movement in relative clauses only, following Huang, 1982a), and Italian (no *wh*-island effect).¹⁵ Crucially, all L1 groups treated the two types of violations differentially, but, most importantly, Indonesian and Chinese speakers rejected strong violations 87% and 76% of the time, respectively, whereas the rejection rate for weak violations was significantly lower (42% and 38%, respectively).

As Schwartz and Sprouse (2000) pointed out, this difference in degree of ungrammaticality constitutes a poverty-of-the-stimulus situation: Neither strong nor weak violations are heard in the input and neither kind is possible in the L1, yet learners showed sensitivity to this distinction. In other words, L2 learners' superior performance on strong violations supports the claim for UG access and suggests sensitivity to degrees of grammaticality, as realized in terms of barriers. However, proponents of the FDH (e.g., Bley-Vroman, 1989, 1990) might legitimately claim that L2 learners should perform better on weak violations than they actually do: After all, extraction from weak islands is still ungrammatical, although to a lesser extent than extraction from strong islands.

If weak violations are prohibited by the same mechanisms as strong violations (although involving fewer barriers), the fact that so many L2 learners have problems in even identifying weak violations as ungrammatical would seem to be unexpected on a UG account. Next, we will suggest that an alternate formulation of island constraints makes better sense theoretically and provides better empirical coverage of the L2 data. This alternate formulation no longer covers some of the original constraints subsumed by subadjacency, making it less surprising that many L2 learners fail to recognize certain sentence types as ungrammatical.

ISLANDS VERSION 3: MINIMALISM

The *Barriers* system (Chomsky, 1986a) did not survive long, on theoretical grounds. In Chomsky's (1995) words, this system lacked "the generality and clarity that we would hope to find in an explanatory theory of language" (p. 89). When the *Barriers* system was challenged by Minimalism (Chomsky, 1995), the idea was that the highly articulated theoretical machinery associated with a "rich UG" (Bley-Vroman, this issue, p. 182) should be minimized as much as possible: Ideally, a principled explanation would be found for any previously stipulated components of UG. In particular, UG principles should not be formulated as arbitrary substantive statements but should be derived from general considerations of economy, interface conditions, and so forth. Although Chomsky's approach (1986a) to island phenomena has been abandoned, insights about what L2 learners might be doing are, in our opinion, not necessarily lost.

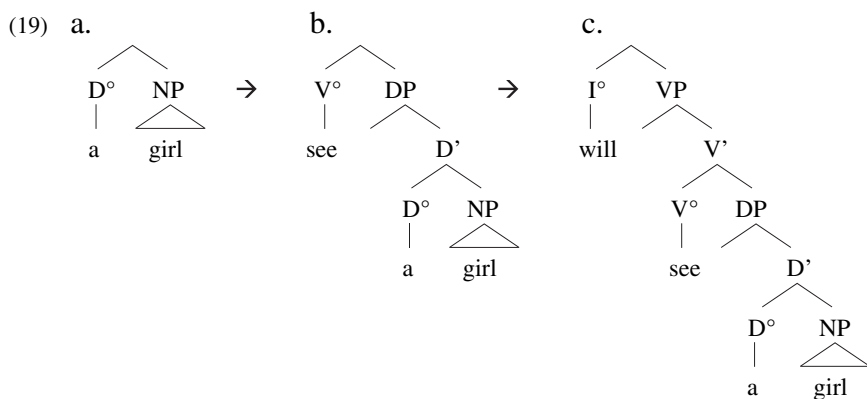
Based, in part, on the CED (Huang, 1982a), the formulation in (18) has been proposed as a universal constraint on extraction that prohibits movement out of noncomplements crosslinguistically (Horvath & Siloni, 2003; Müller, 2007; Nunes & Uriagereka, 2000; Sabel, 2002). This formulation unifies the fact that subjects and adjuncts are universally islands (e.g., in contrast to *wh*-islands).¹⁶

- (18) Revised CED: *Wh*-movement cannot take place out of a structural noncomplement.¹⁷

This revised constraint on extraction is very basic and relies only minimally on any particular theoretical assumption (e.g., government is not involved). UG-access models that assume the revised CED rather than the *Barriers* system (Chomsky, 1986a) would still predict differential treatment of strong versus weak violations by L2 learners. However, weak violations will now have to stem from independent factors: Thus, poor performance on weak violations is not necessarily unexpected.

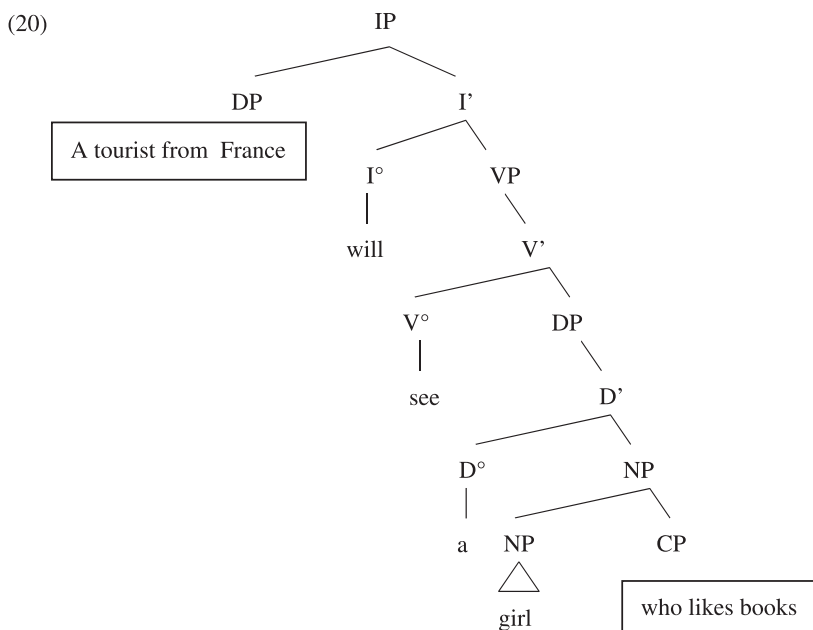
Because the revised CED is still an arbitrary substantive statement, we will briefly outline how this version of island constraints can be derived from more general principles and on more solid conceptual grounds, along the lines of Uriagereka's (1999) multiple spell-out system. The starting point of the multiple spell-out system was a linearization requirement. Although the computational system requires hierarchical structure (visualized as trees in the linguistic tradition), human articulatory and auditory systems are unable to produce and perceive hierarchical relations; instead, these systems can only deal with linear sequences of words. Because the output of the grammar must interface with the articulatory and perceptual systems, hierarchical phrase structure has to be transformed into a flat sequence of terminal nodes (Nunes & Uriagereka, 2000).

Another assumption is that derivation proceeds in a bottom-up fashion. In the case of complements, already-assembled constituents are merged with the heads that select them, which is illustrated in (19). Structure is built up recursively, so that a phrase (the NP *girl*) is first merged with its selecting head (the determiner [D] *a*) and is then completed into a new phrase (the DP *a girl*) that is subsequently merged with its selecting head (the V *see*), which is then completed into a yet larger phrase (the VP *see a girl*) and so forth. In contrast, subjects and adjuncts are phrases that are not selected by heads, and, therefore, they must be assembled in parallel with the pivotal complement structure, in order to be "plugged in" (Nunes & Uriagereka, 2000, p. 23) later where they belong.



The late insertion of subjects and adjuncts makes their internal structure invisible to the computational system; that is, no phrase can be extracted out of a subject or an adjunct. Returning to linearization, once a phrase leaves its derivational workspace, it is transformed into a flat

sequence of words: Because subjects and adjuncts get assembled separately from the rest of the structure (i.e., in a separate derivational workspace), they are necessarily linearized before they are merged into the main structure, as illustrated in (20). In consequence, the computational system can still see subjects and adjuncts and manipulate them if needed (e.g., subjects and adjuncts can be moved as chunks). Crucially, it cannot see into subjects and adjuncts: Once they have undergone linearization, the constituent parts of subjects and adjuncts are no longer there and thus cannot be extracted.



In short, strong islands are covered by the revised CED, itself derived from more general principles operative in every language. Crucially, the multiple spell-out system remains complex and exclusive to language, thus unable to be recovered by domain-general reconstruction restricted to “obvious large-scale characteristics” (Bley-Vroman, 1989, p. 52). In other words, if native speakers of *wh*-in-situ languages respect the revised CED in the acquisition of a L2 that employs *wh*-movement, this suggests that SLA must indeed be “integrat[ed] into a specialized language faculty” (Bley-Vroman, this issue, p. 180).

In sum, the L2 data are largely consistent with the claim that L2 learners observe the revised CED: The ungrammaticality of weak islands is now assumed to be derived from independent factors. Hence, the fact that L2 learners do less well on weak island violations is not surprising.

WEAK ISLANDS

Islands not accounted for under Uriagereka's (1999) multiple spell-out system (viz. *wh*-islands and clausal complements of nouns) must receive an alternate explanation. First, at least some (Stowell, 1981)—or perhaps all (Grimshaw, 1990)—clausal complements of nouns behave like adjuncts according to many diagnostics, so that their status (as either complements or adjuncts) is not obvious. Second, ungrammaticality of extraction out of *wh*-islands in English can be attributed to parametric variation related to either (un)availability of an extra landing site (Reinhart, 1981) or the nature of an additional barrier (Chomsky, 1986a, building on Rizzi, 1982). At least in the case of *wh*-islands, determining the status of extractions out of weak islands appears to require actual input-driven learning of certain properties of the target grammar rather than reliance on principles already operative in the L1 or available through UG.

In the case of clausal complements extracted out of nouns, Chomsky (1986a), among others, noted that such extraction results in milder ungrammaticality compared with extraction out of relative clauses, as illustrated by the comparison of (21a) and (21b) with (21c).

- (21) a. ?*Which book did John hear [a rumor [that you had read t]]?
 b. ?Which book did John announce [a plan [(for you) to read t]]?
 c. *Which book did John meet [a child [who read t]]?

(Chomsky, 1986a, p. 35)

Recall that relative clauses are not complements, but adjuncts that modify the head noun; hence, the ungrammaticality of (21c) is accounted for by either the *Barriers* framework (due to a strong island; Chomsky, 1986a) or under the revised CED (as a noncomplement). The ungrammaticality of extraction from noun complements as illustrated in (21a) and (21b), in contrast, is not readily explained within the *Barriers* system because *wh*-movement out of a clausal complement to a noun crosses no barrier. In other words, such sentences should be fully grammatical, but this is not the case.

In an attempt to account for the intermediate status of such extractions, Chomsky (1986a) speculated that nouns never truly govern their complement position for several reasons. First, nouns assign an oblique inherent case (Chomsky, 1986b), which results in the insertion of the dummy preposition *of* when the complement of the N is NP, as illustrated in (22). Although *of*-insertion does not occur with clausal complements, as illustrated in (23), there may still be something deficient about the intended government relation between the N and its designated complement. Second, deletion of the complementizer *that*, permitted in the case of complements of verbs, as illustrated in (24), is not

licensed for sentential complements of nouns, as illustrated in (25), which suggests that N (in contrast to V) cannot properly govern a null C (Stowell, 1981).

- (22) a. **the* [*destruction* [*the city*]]
 b. *the* [*destruction* [*of the city*]] (Chomsky, 1986b, p. 200)
- (23) a. *Paul's* [*explanation* [*that he was temporarily insane*]].
 b. **Paul's* [*explanation of* [*that he was temporarily insane*]]. (Stowell, p. 193)
- (24) *He claimed* [(*that*) *Bill had left the party*].
- (25) a. **I distrust the* [*claim* [*Bill had left the party*]].
 b. *I distrust the* [*claim* [*that Bill had left the party*]]. (Stowell, p. 398)

This failure of government explains why extraction is not possible: Complements of Ns have a different status from complements of other lexical categories and share with adjuncts the property of not being governed. At the same time, extraction out of N complements is not as bad as extraction out of relative clauses (which are true adjuncts), a fact that remains unexplained.

Related proposals are elaborated in Stowell (1981) and Grimshaw (1990). Stowell distinguished between finite complements of Ns, as illustrated in (21a), and infinitival complements, as illustrated in (21b), and argued that only infinitival complements are true complements.¹⁸ Stowell suggested that finite CPs are modifiers, a claim independently motivated by the fact that Ns with such CPs are not generally eventive—compare *claim*, which takes a finite clause, with *attempt*, which takes an infinitival clause—and therefore have no theta-role to assign. This distinction between infinitival and finite N complements predicts that extraction out of infinitival complements is better than extraction out of finite CPs—a prediction that is supported, as can be seen in the comparison of (21a) and (21b). Nevertheless, extraction out of infinitival complements of Ns is not as good as extraction out of CP complements of Vs (a contrast that remains to be accounted for).

A somewhat different analysis was proposed by Grimshaw (1990), who argued that neither infinitival nor finite N complements behave like true arguments. First, Ns are defective theta-markers, in that they can only transmit their theta-roles through prepositions (an option available for NP complements but not for CPs). Second, Ns that take CPs lack an aspectual dimension and thus act as result nominals (see Stowell, 1981, for finite clauses) or simple event nominals but not as complex event nominals.¹⁹ Furthermore, clausal complements of Ns are never obligatory, although true (non-CP) complements of Ns become obligatory when the event reading is forced; as illustrated in (26).

- (26) a. **The frequent expression is desirable.*
 b. *The frequent expression of one's feelings is desirable.* (Grimshaw, p. 50)

Grimshaw suggested, in short, that CP complements of Ns are mixed in nature, in that they behave like modifiers but still respect selectional properties.

In sum, there appears to be a consensus that clausal complements of Ns are in some way special. Although the current generative framework does not offer a three-way distinction (complement, adjunct, and some other category), such a distinction might in fact be needed.²⁰ CP complements of Ns have a special status: They are neither true complements nor straightforward adjuncts. This explains why the revised CED, which makes crucial reference to noncomplements, neither fully bans movement in such cases nor readily allows it.

Second language learners' depressed performance on extractions from N complements, then, may in fact reflect a sensitivity to their mixed status. Indeed, what is odd on this account is the overall tendency of native speakers to reject extractions out of clausal complements of Ns rather than showing the same ambivalence as L2 learners. However, some native speakers do accept such extractions to a certain degree; for example, Schachter (1989) noted that native speakers performed relatively poorly on the part of the test that involved N complements. Native controls reported by Martohardjono (1993) as well as by Johnson and Newport (1991) also showed a tendency to accept extractions from N complements.

Turning now to *wh*-islands, grammatical long-distance *wh*-movement is cyclical, proceeding by means of a series of local movements. In English, only [Spec, CP] is available as an intermediate landing site for the *wh*-phrase. This means that extraction of a *wh*-phrase like *what* out of an embedded question (a *wh*-island), as illustrated in (27), is not possible because it violates the cyclicity requirement, as the embedded [Spec, CP] is already occupied by *to whom*.

- (27) **What_i did you wonder* CP[[*to whom*]_k IP[*John gave t_i t_k]]]?
 (Chomsky, 1986a, p. 36)*

In contrast, if a language allows for more than one landing site, *wh*-movement out of an embedded question can take place cyclically, as has been argued for Hebrew by Reinhart (1981). The sentence in (28) is grammatical, in contrast to its English equivalent; the CP is arguably split into two layers, which allows the *wh*-phrase to escape.

- (28) [*eize sefer*]_i *shaxaxta* CP²[[*mi-mi*]_k CP¹[*t_i IP[kibalta t_i t_k]]]]?
 which book you forgot from whom you got
 "Which book is such that you forgot from whom you got it?"
 (Reinhart, p. 533)*

Italian also allows extraction out of *wh*-islands, although less freely than Hebrew. In particular, whereas Hebrew licenses extraction out of any number of embedded questions, Italian counterparts of such sentences are considerably degraded (Rizzi, 1982). In brief, Reinhart (1981) captured this difference (and other points of divergence) between Hebrew and Italian with the proposal that languages differ in how many landing sites they permit; this difference must be acquired by learners.

This account differs from the explanation offered by Rizzi (1982), who advanced a parameterized formulation of subjacency. He proposed that, in Italian, CP rather than IP was a bounding node (in addition to DP). Chomsky (1986a) restated Rizzi's parameterization analysis in terms of barriers; specifically, Chomsky proposed an additional inherent barrier (viz. the most deeply embedded tensed IP in English or CP in Italian).²¹ What Chomsky intended to capture, in addition to parameterization, is the fact that extractions out of infinitival embedded questions like (29) are in fact acceptable for many speakers in English and thus cannot involve crossing more than one barrier (the CP, which is a barrier by inheritance). Note that this is reminiscent of the situation with clausal N complements, which are also weak islands and become even easier to extract from when movement takes place out of an infinitival embedded clause.

(29) ?Which car did John tell you [how to fix t]?

(Chomsky, p. 37)

Interestingly, Chomsky (1986a) remarked that many native speakers of English show the Italian judgments and suggested that "this should not be surprising with regards to low-level parametric variation that receives little support in evidence available to the language learner" (p. 37). In other words, there might be too little positive evidence for even native English speakers to converge on what would count as representative of English grammar. Indeed, data from native speakers in Martohardjono (1993) and Johnson and Newport (1991), among others, reflected depressed performance on *wh*-island violations, which suggests that these violations are occasionally acceptable in English. The immediate corollary, then, is that it is inappropriate to interpret L2 learners' failure to reject extractions out of *wh*-islands in English as evidence for a UG deficit. Given that L2 learners encounter even less positive evidence in the target language than L1 learners, if L1 learners cannot converge on the correct value when such low-level parametric variation is concerned, L2 learners' failure to reject extractions out of *wh*-islands is irrelevant to the UG-access debate in SLA.

In sum, it appears that determining the status of extractions from *wh*-islands and clausal complements of Ns requires something that goes beyond the revised CED. On the assumption that UG constrains inter-language grammars, the revised CED should be automatically available

upon switching to the [+*wh* movement] value of the *wh*-movement parameter; therefore, L2 learners should be able to immediately recognize extractions out of noncomplements (i.e., subjects and adjuncts) as impossible. This is largely borne out: L2 learners are quite accurate on extractions out of noncomplements. However, L2 learners cannot be expected to be correspondingly accurate at recognizing milder violations (which involve clausal complements of Ns and *wh*-islands) because these milder violations are not straightforwardly ruled out by the same mechanism. Indeed, the fact that many L2 speakers perform largely at chance on weak violations only serves to highlight the fact that L2 learners are highly sensitive to what the target grammar (and UG) truly prohibits.

DISCUSSION: WHERE ARE WE NOW?

Changes to theoretical accounts of island effects have had consequences for theories of SLA and have affected the debate between the FDH and UG-access approaches. The original island constraints (Chomsky, 1964, 1973; Ross, 1967) were brought together under subadjacency, which defined islands in terms of bounding nodes. It is this account of islands that attracted a considerable amount of L2 research. Several researchers (e.g., Schachter, 1989, 1990) have reported that L2 learners showed low accuracy, often at chance levels, on subadjacency violations, failing to recognize them as ungrammatical. These researchers concluded that UG no longer constrains adult grammars, which supports the FDH.

What remained unexplained (although not necessarily unremarked) in these early accounts was that performance often differed on the various subtypes of violations; L2 learners demonstrated considerably greater accuracy in the recognition of violations that involved extraction out of adjuncts or subjects. As pointed out by Martohardjono (1993), this phenomenon was amenable to an account in terms of Chomsky's (1986a) *Barriers* system, which formalized the concept of degrees of grammaticality, depending on how many barriers had been crossed. With this system, empirical coverage improved: It became possible to account for the observation that L2 learners are more sensitive to strong islands, something that could not have been learned on the basis of L2 input or arrived at on the basis of the L1, which supported claims that interlanguages are UG-constrained. It is important to note that, under both subadjacency and (to a lesser degree) the *Barriers* framework, constraints on extraction took the form of universals somehow inactive in *wh*-in-situ languages. L2 learners' sensitivity to strong islands thus implicated UG access beyond the L1. However, this system was, in turn, superseded by accounts consistent with Minimalism,

accounts that continue to emphasize the importance of subjects and adjuncts (i.e., noncomplements) as islands but that derive the effects from more general (although still domain-specific) computational principles.

Most of the principles previously assumed to constitute a rich UG will, ideally, turn out to be byproducts of more general computational (but nevertheless linguistic) mechanisms, which are active in every human language. If interlanguage grammars are constrained by such mechanisms, L2 learners' sensitivity to strong islands (i.e., the revised CED) can indeed be accounted for in terms of transfer, albeit rather sophisticated transfer. Indeed, it no longer makes sense to test the FDH and UG-access models against each other by looking for some universal principle inactive in a particular L1 but active in the L2. Rather, it now appears to be impossible to distinguish between the effects of the L1 and of UG. In particular, Uriagereka's (1999) multiple spell-out system must be operative in every language. All languages, including languages without *wh*-movement, have subjects and adjuncts. The general mechanism that renders subjects and adjuncts inaccessible to further internal manipulation, including extraction, will therefore be universal. What has to be acquired is the fact that closing off subjects and adjuncts affects *wh*-movement possibilities. Once native speakers of [*-wh* movement] languages reset the *wh*-movement parameter to [*+wh* movement], the differential treatment of ungrammatical extractions out of complements and noncomplements can be explained via a subtle form of L1 transfer: No *wh*-extraction is possible out of an already closed-off domain. However, crucially, this is not the same as the original FDH proposal that the L1 can be used to reconstruct UG. On the contrary, the principles that underlie constraints on movement are too linguistic in nature to allow domain-general reconstruction; rather, computations and mechanisms still active in the L1 are also relevant for the L2, so no reconstruction is even required.

In conclusion, the investigation of island effects has resulted in changes in what counts as universal, in theoretical accounts of the phenomena in question, and in what could be L1 based. Despite these changes, we are nevertheless dealing with the SLA of linguistic properties that constitute a poverty of the stimulus. The behavior of L2 learners in this domain is consistent with the claim that learners are restricted to grammars that fall within the range permitted by UG. In other words, changes in linguistic theory do not require the abandonment of the observation that L2 grammars are UG-constrained. However, assuming the operation of the same computational principles in the grammars of all languages, it is now a moot point as to whether it makes sense to look for a pure UG explanation (Dekydtspotter et al., 1997; Hale, 1996; White, 1995). Where does this leave the FDH? Is it indeed possible for L1 and L2 acquisition to be fundamentally different?

NOTES

1. Neither Johnson and Newport (1991) nor Schachter (1989, 1990) specifically identified themselves with the FDH as such.

2. In some cases, additional assumptions have to be made to account for ungrammaticality.

3. The claim that *wh*-in-situ languages involve no *wh*-movement is not straightforward. If the rationale behind *wh*-movement is to allow *wh*-phrases to be interpreted as operators that bind variables, then *wh*-in-situ languages might be expected to employ an equivalent strategy at some level. Indeed, Chomsky (1973) and Huang (1982a) argued that *wh*-phrases in *wh*-in-situ languages undergo LF (or covert) *wh*-movement, analogous to overt *wh*-movement. However, Huang argued that LF *wh*-movement is immune to island constraints (see also Baker, 1970; Chomsky). Although this appears to be true of Chinese, which exhibits no subadjacency effects in questions, Japanese (Nishigauchi, 1986) and Korean (Lee, 1982; Nishigauchi, 1990) respect *wh*-islands.

4. Negative evidence is presumably either nonexistent or noneffective (Schwartz, 1993). In any case, it is inconceivable that L2 explicit classroom instruction includes information about the kinds of ungrammaticality found in (3b)–(7b).

5. Many studies have explored the SLA of a *wh*-movement language (usually English) by native speakers of *wh*-in-situ languages. Only a few studies have investigated the other direction. These studies have mainly focused on interpretations of *wh*-phrases and scope effects in Japanese, Korean, and Mandarin rather than on island effects (Choi & Lardiere, 2006a, 2006b; Umeda, 2008; Yuan, 2008).

6. Li (1998) did not in fact provide a definition of subadjacency, but this version of island constraints is implicit in her analysis.

7. Other early studies focused on the parameterization of bounding nodes (e.g., White, 1985, 1988).

8. Schachter (1989, 1990) also included Indonesian, and Schachter (1990) included Dutch as well. These languages have been omitted from this discussion.

9. For a discussion of problems with the sentences used by some of these studies, see White (2003).

10. Scrambling (or free word order) refers to the phenomenon of (arguably) optional noncanonical word orders allowed in some languages. The exact nature of scrambling (stylistic movement, base generation, A-movement, or A'-movement) has long been a matter of debate (see Hawkins & Hattori, 2006).

11. *Wh*-scrambling is not equivalent to *wh*-movement; *wh*-scrambling is optional and is not feature driven (Saito & Fukui, 1998). However, according to Saito (1985) for Japanese and Kim (1989) for Korean, scrambling is in fact subject to subadjacency.

12. Lexical categories (such as N, V, and preposition) are distinguished from functional categories (such as I and C). Note however that CED does not account for the ungrammaticality of extractions out of clausal complements of Ns (a lexical category) unless additional assumptions are made.

13. The ECP is a separate constraint, which places conditions on empty categories rather than on movement as such. The combination of ECP and subadjacency violations often makes sentences even more ungrammatical than subadjacency violations alone. Some L2 studies looked at the ECP as well as subadjacency (e.g., Bley-Vroman et al., 1988; Martohardjono, 1993; White, 1988); in some cases, the cumulative effect of both constraints was examined (Li, 1998; Martohardjono; Uziel, 1993).

14. In fact, if no additional assumption is made (Chomsky, 1986a), extractions from N complements become grammatical under this system.

15. For speakers of L1s with [+*wh* movement], see Uziel (1993), who presented similar results from speakers of Hebrew.

16. However, see Stepanov (2007), who argued that the subject condition is not universal.

17. The term structural noncomplement is used in recognition of the fact that extraction from VPs and IPs is permitted, even though these phrasal categories are not theta-marked by a lexical head but are complements of the functional heads I and C, respectively.

18. Stowell (1981) argued that case-marking must take place for an argument to be visible for theta-assignment (following Chomsky, 1981). Based on diagnostics such as topicalization, order of verbal complements, and passivization, Stowell suggested that infinitival CPs are intrinsically case-marked. In contrast, finite CPs cannot be case-marked: When they are complements of Vs, finite CPs move out of the V', and their trace is case-marked by the theta-assigning V. Adopting the idea that Ns do not assign case, Stowell concluded that only infinitival CPs can be true complements of Ns: As they have inherent case, infinitival CPs do not need to get case from the N.

19. Grimshaw (1990) crucially assumed that both the aspectual and thematic dimensions of a predicate are obligatory in argument structure; thus, the lack of an aspectual dimension necessarily implies the lack of argument structure. The diagnostics for aspectual dimensions that Grimshaw used include modification by aspectual adjectives like *frequent* and *constant* as well as the ungrammaticality of pluralization and predicative usage: All nominals consistently fail these tests when they appear with sentential complements.

20. The assumption here is that subjects and modifiers are collapsed under adjuncts, as in Kayne (1984).

21. Such an account does not work even within the *Barriers* system (Chomsky, 1986a). See Chomsky for discussion of some immediate problems.

REFERENCES

- Baker, C. L. (1970). Notes on the description of English questions: The role of an abstract question morpheme. *Foundations of Language*, 6, 197–219.
- Bley-Vroman, R. (1989). What is the logical problem of foreign language learning? In S. M. Gass & J. Schachter (Eds.), *Linguistic perspectives on second language acquisition* (pp. 42–68). New York: Cambridge University Press.
- Bley-Vroman, R. (1990). The logical problem of foreign language learning. *Linguistic Analysis*, 20, 3–49.
- Bley-Vroman, R., Felix, S., & Ioup, G. (1988). The accessibility of Universal Grammar in adult language learning. *Second Language Research*, 4, 1–32.
- Cattell, R. (1976). Constraints on movement rules. *Language*, 52, 18–50.
- Choi, M.-H., & Lardiere, D. (2006a, April). *Acquiring the interpretation of wh-expressions in Korean as a second language*. Paper presented at Generative Approaches to Second Language Acquisition 8 (GASLA 8), Banff, Canada.
- Choi, M.-H., & Lardiere, D. (2006b). The interpretation of *wh*-in-situ in Korean second language acquisition. In A. Belletti, E. Bennati, C. Chesi, E. Di Domenico, & I. Ferrari (Eds.), *Proceedings of the 2005 Conference on Generative Approaches to Language Acquisition (GALA 2005)* (pp. 125–136). Cambridge: Cambridge Scholars Press.
- Chomsky, N. (1964). *Current issues in linguistic theory*. Berlin: Mouton de Gruyter.
- Chomsky, N. (1973). Conditions on transformations. In S. R. Anderson & P. Kiparsky (Eds.), *A festschrift for Morris Halle* (pp. 232–286). New York: Holt, Rinehart, & Winston.
- Chomsky, N. (1981). *Lectures on government and binding*. Dordrecht: Foris.
- Chomsky, N. (1986a). *Barriers*. Cambridge, MA: MIT Press.
- Chomsky, N. (1986b). *Knowledge of language: Its nature, origin and use*. New York: Praeger.
- Chomsky, N. (1995). *The Minimalist Program*. Cambridge, MA: MIT Press.
- Crain, S., & Lillo-Martin, D. (1999). *An introduction to linguistic theory and language acquisition*. Oxford: Blackwell.
- Dekydtspotter, L., Sprouse, R. A., & Anderson, B. (1997). The interpretive interface in L2 acquisition: The process-result distinction in English-French interlanguage grammars. *Language Acquisition*, 6, 297–332.
- Grimshaw, J. (1990). *Argument structure*. Cambridge, MA: MIT Press.
- Hale, K. (1996). Can UG and the L1 be distinguished in L2 acquisition? *Brain and Behavioral Sciences*, 19, 728–730.
- Hawkins, R. (2001). The theoretical significance of Universal Grammar in second language acquisition. *Second Language Research*, 17, 345–367.

- Hawkins, R. (2005). Revisiting *wh*-movement: The availability of an uninterpretable [wh] feature in interlanguage grammars. In L. Dekydtspotter, R. A. Sprouse, & A. Liljestr nd (Eds.), *Proceedings of the 7th Generative Approaches to Second Language Acquisition Conference (GASLA 2004)* (pp. 124–137). Somerville, MA: Cascadia Press.
- Hawkins, R., & Chan, Y.-H.C. (1997). The partial availability of Universal Grammar in second language acquisition: 'The failed functional features hypothesis.' *Second Language Research*, 13, 187–226.
- Hawkins, R., & Hattori, H. (2006). Interpretation of English multiple *wh*-questions by Japanese speakers: A missing uninterpretable feature account. *Second Language Research*, 22, 269–301.
- Horv th, J., & Siloni, T. (2003). Against the little-*v* hypothesis. In Y. N. Falk (Ed.), *Proceedings of the Israel association for theoretical linguistics (IATL)*, 19, 1–15. Retrieved November 17, 2007, from <http://linguistics.huji.ac.il/IATL/19/HorvathSiloni.pdf>.
- Huang, C.-T. (1982a). *Logical relations in Chinese and the theory of grammar*. Unpublished doctoral dissertation, MIT, Cambridge, MA.
- Huang, C.-T. (1982b). Move *wh* in a language without *wh*-movement. *The Linguistic Review*, 1, 369–416.
- Johnson, J., & Newport, E. (1991). Critical period effects on universal properties of language: The status of subadjacency in the acquisition of a second language. *Cognition*, 39, 215–258.
- Kayne, R. (1984). *Connectedness and binary branching*. Dordrecht: Foris.
- Kim, S. (1989). *Wh*-phrases in Korean and Japanese are QPs. *MIT Working Papers in Linguistics*, 11, 119–138.
- Lee, H.-S. (1982). *Asymmetry in island constraints in Korean*. Unpublished manuscript, University of California Los Angeles.
- Li, X. (1998). Adult L2 accessibility to UG: An issue revisited. In S. Flynn, G. Martohardjono, & W. O'Neil (Eds.), *The generative study of second language acquisition* (pp. 232–286). Mahwah, NJ: Erlbaum.
- Martohardjono, G. (1993). *Wh-movement in the acquisition of a second language: A cross-linguistic study of three languages with and without overt movement*. Unpublished doctoral dissertation, Cornell University, Ithaca, NY.
- Martohardjono, G., & Gair, J. (1993). Apparent UG inaccessibility in second language acquisition: Misapplied principles or principled misapplications? In F. Eckman (Ed.), *Confluence: Linguistics, L2 acquisition and speech pathology* (pp. 79–103). Amsterdam: Benjamins.
- M ller, G. (2007, April). *On deriving CED effects from the PIC*. Paper presented at the 30th Annual Colloquium of Generative Linguistics in the Old World (GLOW XXX), University of Troms , Norway.
- Nishigauchi, T. (1986). *Quantification in syntax*. Unpublished doctoral dissertation, University of Massachusetts, Amherst.
- Nishigauchi, T. (1990). *Quantification in the theory of grammar*. Dordrecht: Kluwer.
- Nunes, J., & Uriagereka, J. (2000). Cyclicity and extraction domains. *Syntax*, 3, 20–43.
- Reinhart, T. (1981). A second COMP position. In A. Belletti, L. Brandi, & L. Rizzi (Eds.), *Theory of markedness in generative grammar* (pp. 517–557). Pisa: Scuola Normale Superiore di Pisa.
- Rizzi, L. (1982). *Issues in Italian syntax*. Dordrecht: Foris.
- Ross, J. R. (1967). *Constraints on variables in syntax*. Unpublished doctoral dissertation, MIT, Cambridge, MA.
- Sabel, J. (2002). A Minimalist analysis of syntactic islands. *The Linguistic Review*, 19, 271–315.
- Saito, M. (1985). *Some asymmetries in Japanese and their theoretical implications*. Unpublished doctoral dissertation, MIT, Cambridge, MA.
- Saito, M., & Fukui, N. (1998). Order in phrase structure and movement. *Linguistic Inquiry*, 29, 439–474.
- Schachter, J. (1989). Testing a proposed universal. In S. M. Gass & J. Schachter (Eds.), *Linguistic perspectives on second language acquisition* (pp. 73–88). New York: Cambridge University Press.
- Schachter, J. (1990). On the issue of completeness in second language acquisition. *Second Language Research*, 6, 93–124.

- Schwartz, B. D. (1993). On explicit and negative data effecting and affecting competence and linguistic behavior. *Studies in Second Language Acquisition*, 15, 147–163.
- Schwartz, B. D., & Sprouse, R. A. (2000). When syntactic theories evolve: Consequences for L2 acquisition research. In J. Archibald (Ed.), *Second language acquisition and linguistic theory* (pp. 156–186). Oxford: Blackwell.
- Sportiche, D. (1981). Bounding nodes in French. *The Linguistic Review*, 1, 219–246.
- Stepanov, A. (2007). The end of CED? Minimalism and extraction domains. *Syntax*, 10, 80–126.
- Stowell, T. (1981). *Origins of phrase structure*. Unpublished doctoral dissertation, MIT, Cambridge, MA.
- Umeda, M. (2005). *Wh-movement in Japanese-English interlanguage: Evidence from scope and reconstruction*. In A. Brugos, M. Clark-Cotton, & S. Ha (Eds.), *Proceedings of the 29th Annual Boston University Conference on Language Development* (pp. 616–626). Somerville, MA: Cascadilla Press.
- Umeda, M. (2006). *Wh-movement in L2 grammars: Evidence for parameter resetting*. In K. U. Deen, J. Nomura, B. Schulz, & B. D. Schwartz (Eds.), *Proceedings of the Inaugural Conference on Generative Approaches to Language Acquisition - North America* (pp. 389–400). Storrs, CT: University of Connecticut Occasional Papers in Linguistics.
- Umeda, M. (2008). *Second language acquisition of Japanese Wh-constructions*. Unpublished doctoral dissertation, McGill University, Montreal.
- Uriagereka, J. (1999). Multiple spell-out. In S. Epstein & N. Hornstein (Eds.), *Working Minimalism* (pp. 251–282). Cambridge, MA: MIT Press.
- Uziel, S. (1993). Resetting Universal Grammar parameters: Evidence from second language acquisition of subadjacency and the empty category principle. *Second Language Research*, 9, 49–83.
- White, L. (1985). The acquisition of parameterized grammars: Subadjacency in second language acquisition. *Second Language Research*, 1, 1–17.
- White, L. (1988). Island effects in second language acquisition. In S. Flynn & W. O'Neill (Eds.), *Linguistic theory in second language acquisition* (pp. 144–172). Dordrecht: Reidel.
- White, L. (1990). Second language acquisition and Universal Grammar. *Studies in Second Language Acquisition*, 12, 121–133.
- White, L. (1992). Subadjacency violations and empty categories in L2 acquisition. In H. Goodluck & M. Rochemont (Eds.), *Island constraints* (pp. 445–464). Dordrecht: Kluwer.
- White, L. (1995). Chasing after linguistic theory: How minimal should we be? In L. Eubank, L. Selinker, & M. Sharwood Smith (Eds.), *The current state of interlanguage: Studies in honor of William E. Rutherford* (pp. 63–71). Amsterdam: Benjamins.
- White, L. (2003). *Second language acquisition and Universal Grammar*. New York: Cambridge University Press.
- White, L., & Juffs, A. (1998). Constraints on *wh*-movement in two different contexts of non-native language acquisition: Competence and processing. In S. Flynn, G. Martohardjono, & W. O'Neil (Eds.), *The generative study of second language acquisition* (pp. 111–129). Mahwah, NJ: Erlbaum.
- Yuan, B. (2008). Discrepancy in English speakers' L2 acquisition of Chinese *wh*-words as existential polarity words: The L1-dependent interface hypothesis. In R. Slabakova, J. Rothman, P. Kempchinsky, & E. Gavruseva (Eds.), *Proceedings of the 8th Generative Approaches to Second Language Acquisition Conference (GASLA 2007)* (pp. 272–284). Somerville, MA: Cascadilla Press.